Predicting power generation based on regional microclimate data

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Outline

- **Introduction to the competition and the data**
- **Methods**
- **Results and analysis**
- **Conclusions**

AI CUP 2024
Competition to predict power generation based on regional microclimate data

- This competition is based on energy technology-related literature, selecting microclimate environmental factors for measurement, collecting data, and using solar panel power generation measurements as the ground truth.

- The content involves providing meteorological data for each device at one-minute intervals. Using the data from the previous day up to 9:00 AM on the prediction day, participants are tasked with forecasting power generation (in watts) every 10 minutes from 9:00 AM to 5:00 PM, resulting in 48 predictions in total. The total error will determine the final score.

- The final ranking is 44th/500.

Data

Data Source and Collection

Measurement locations include three areas in Hualien: Dong Hwa University Main Campus (agricultural zone on an alluvial plain), Meilun Campus (urban area near a stone factory by the Pacific coast), and the Fenglin Solar Development Zone (mountainous area in the East Rift Valley). Measurements were taken at various sites such as rooftops, ground level, ventilated areas, and forest edges, with approximately 20 devices installed.

Data Overview

- **Period**: January to November 2024
- **Frequency**: Per minute
- **Parameters**: Wind speed (m/s), atmospheric pressure (hPa), temperature (°C), relative humidity (%), illuminance (lux), and power generation (W).

Other Data

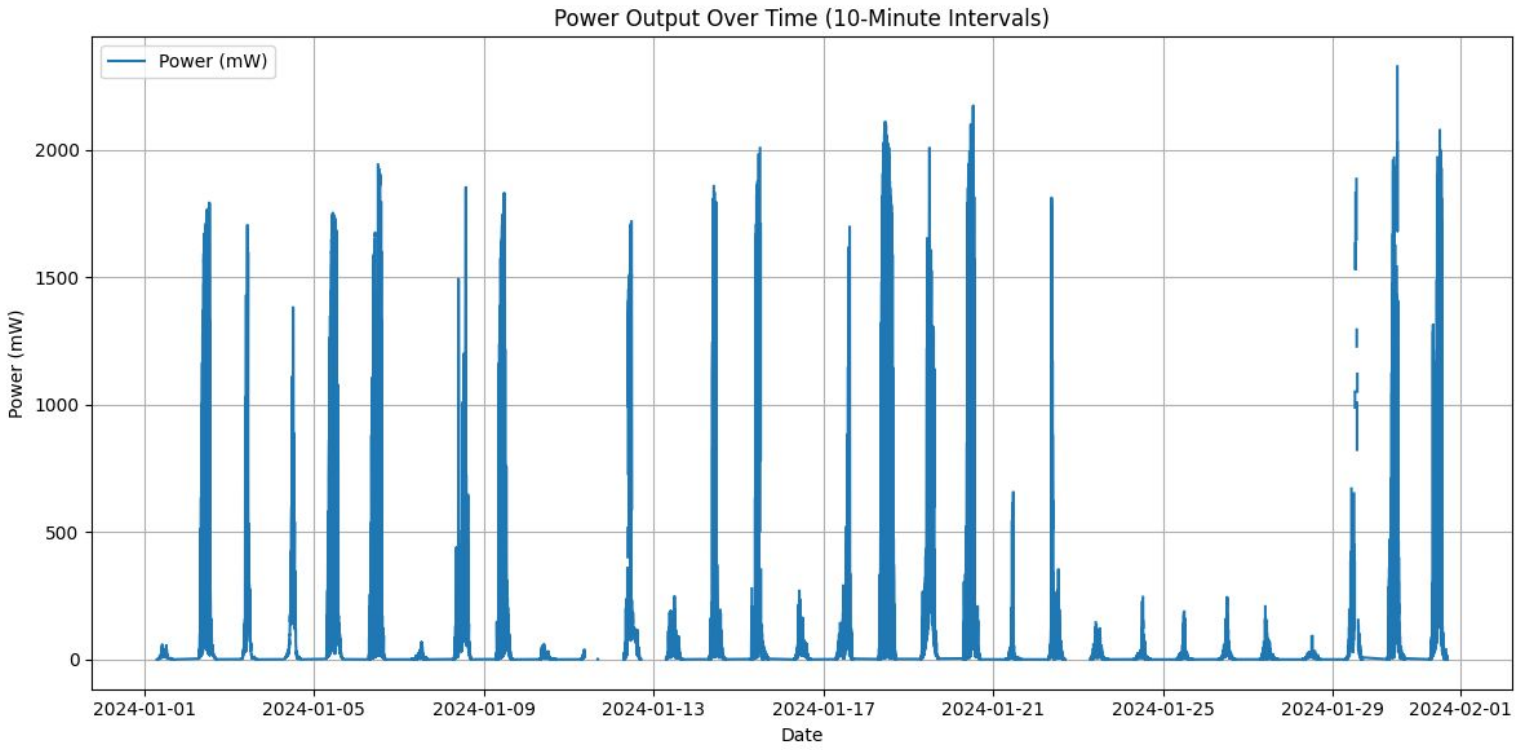
We also incorporated additional data, including daily solar radiation data from the Central Weather Administration and air pollution data (SO_2, CO, O_3, $PM_{2.5}$) from the Ministry of Environment, with a frequency of one record per hour.

Data Preprocessing

- **Sunlight**:
  - Used regression on power generation data below the threshold to estimate sunlight.
  - For values exceeding the threshold, back-calculated sunlight using daily power generation.
- **Wind Speed**:
  - Dropped due to excessive missing values.
- **Location**:
  - Dropped, as location differences were already reflected in the data.
- **Time Interval**:
  - Averaged data into 10-minute intervals.
  - Marked intervals with more than three missing values as NA.
- **External Data**:
  - Merged external datasets after averaging.

```
第1個觀測站的風速缺失比例為：78.03%
第2個觀測站的風速缺失比例為：45.79%
第3個觀測站的風速缺失比例為：69.64%
第4個觀測站的風速缺失比例為：72.01%
第5個觀測站的風速缺失比例為：28.94%
第6個觀測站的風速缺失比例為：31.69%
第7個觀測站的風速缺失比例為：99.58%
第8個觀測站的風速缺失比例為：86.77%
第9個觀測站的風速缺失比例為：87.75%
第10個觀測站的風速缺失比例為：71.84%
第11個觀測站的風速缺失比例為：76.84%
第12個觀測站的風速缺失比例為：91.99%
第13個觀測站的風速缺失比例為：75.42%
第14個觀測站的風速缺失比例為：72.71%
第15個觀測站的風速缺失比例為：77.05%
第16個觀測站的風速缺失比例為：90.74%
第17個觀測站的風速缺失比例為：100.00%
```

Data Visualization



Summary Table

Metrics	Pressure(hpa)	Temperature(°C)	Humidity(%)	Sunlight(Lux)	solar_irad	Power(mW)	date
count	83277	83277	83277	83277	123549	83277	123549
mean	1010.44	30.48	64.95	26675.85	1.43	296.26	2024-04-25 07:12
min	994.21	10.87	11.67	57.5	0	0.01	2024-01-01 07:00
25%	1005.32	23.43	45.75	5679.75	0.45	7.74	2024-03-18 13:20
50%	1009.81	29.75	65.14	12805.83	1.15	39.77	2024-04-26 09:38
75%	1015.01	36.16	85.7	34104.58	2.35	310.57	2024-05-30 08:20
max	2151.94	59.95	100.01	215982.8	3.99	2327.21	2024-09-19 11:44
std	8.49	8.95	24.07	32722.14	1.11	497.03	NaN

Data Visualization

Pressure(hpa)

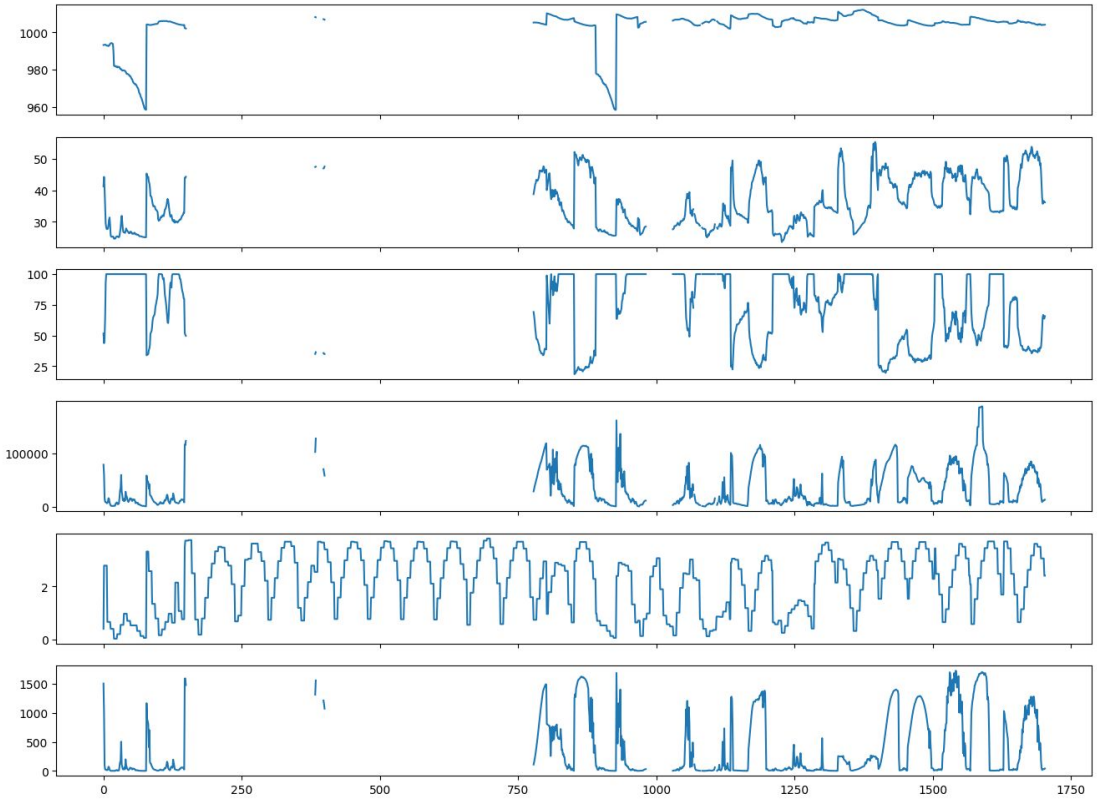
Temperature(℃)

Humidity(%)

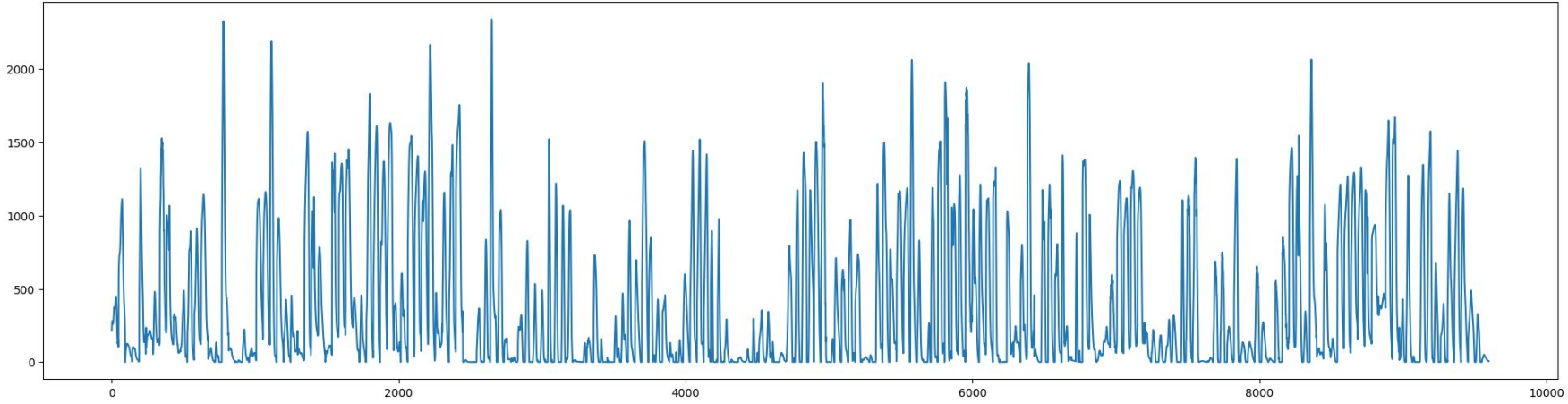
Sunlight(Lux)

Solar_irad

Power(mW)



Prediction Goal



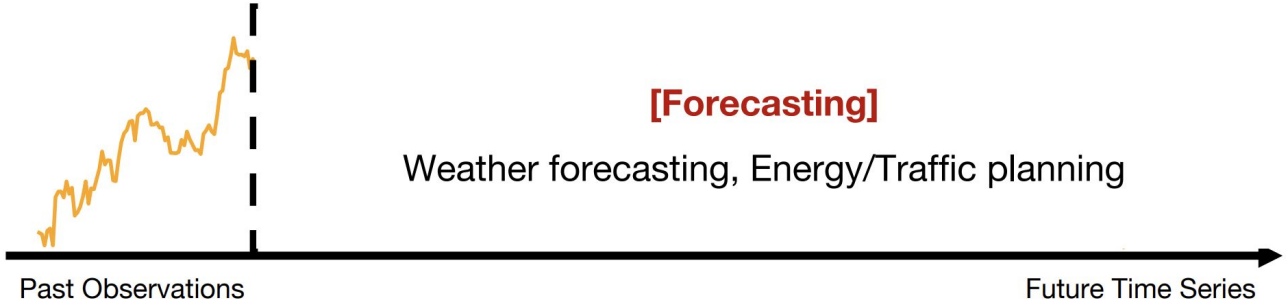
Models

- **Transformer Based**:
  - Autoformer
  - Crossformer
  - Non-stationary transformer
  - Patch-TST
- **CNN Based**
  - TimesNet
  - SCINet
- **RNN Based**
  - Mamba
  - LSTM

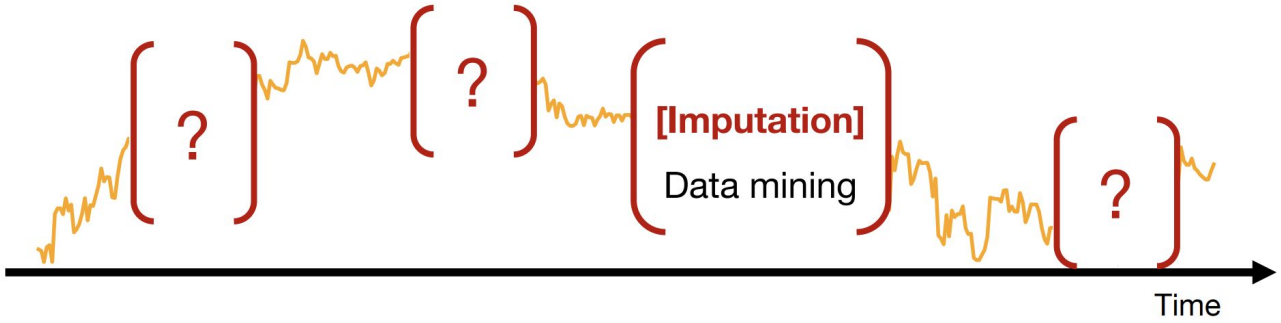
- **Prediction Based Line** : LSTM

Methods

Forecasting: **past** observations

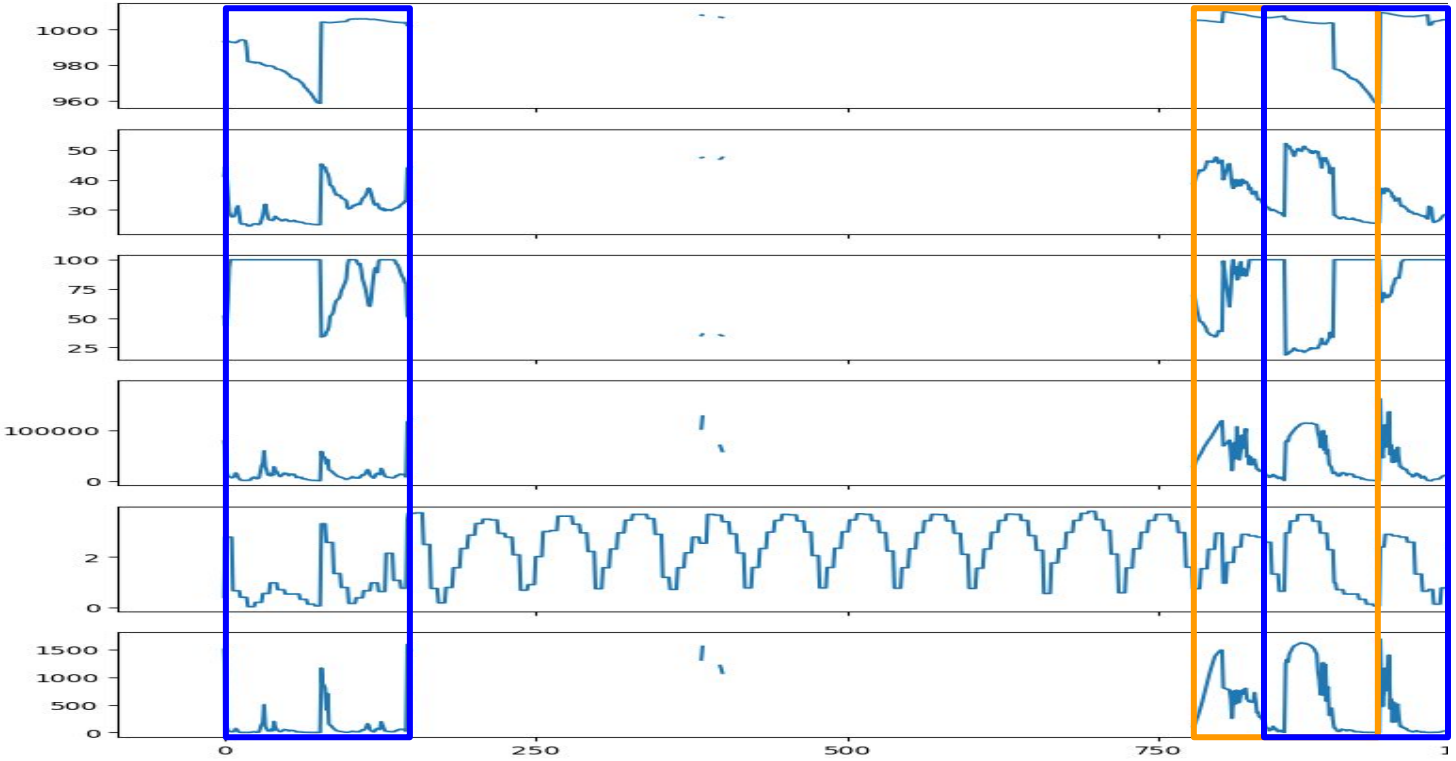


Imputation: **past** and **future** observations

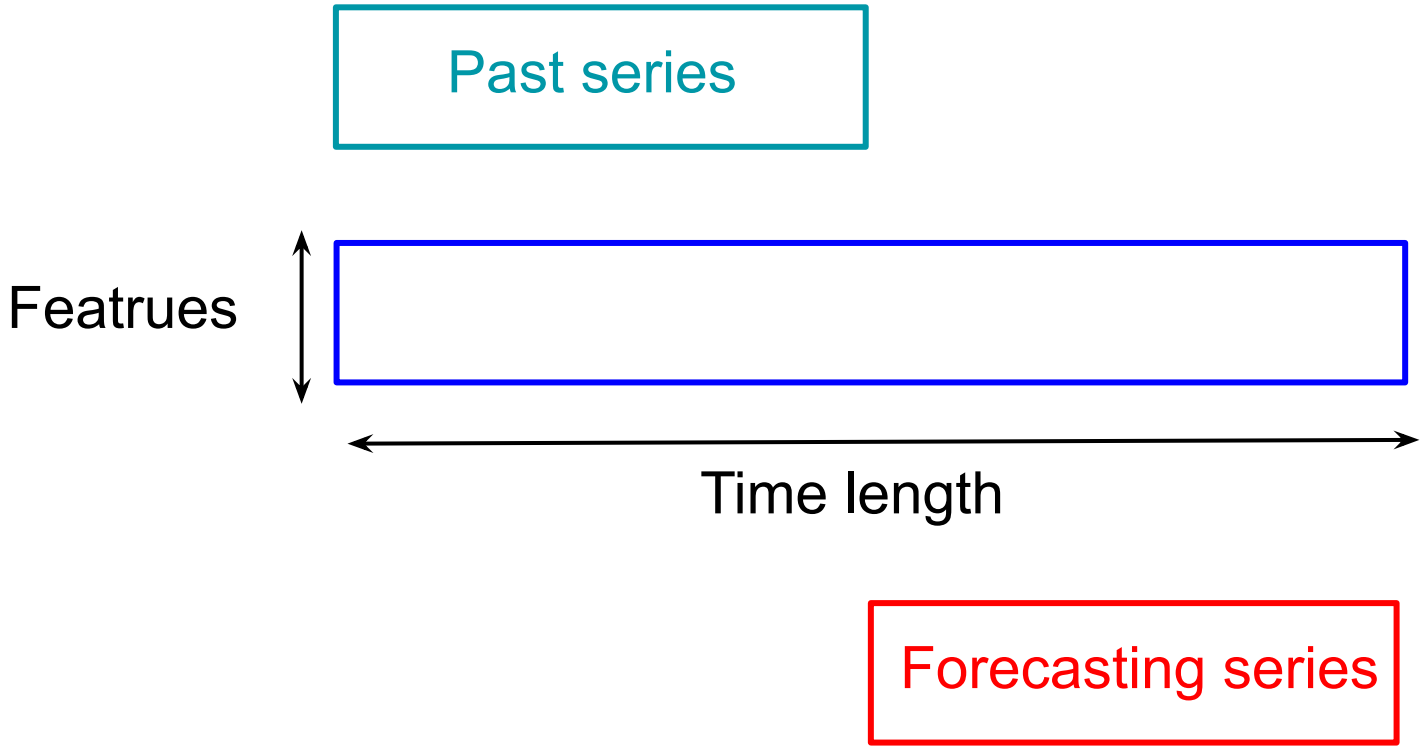


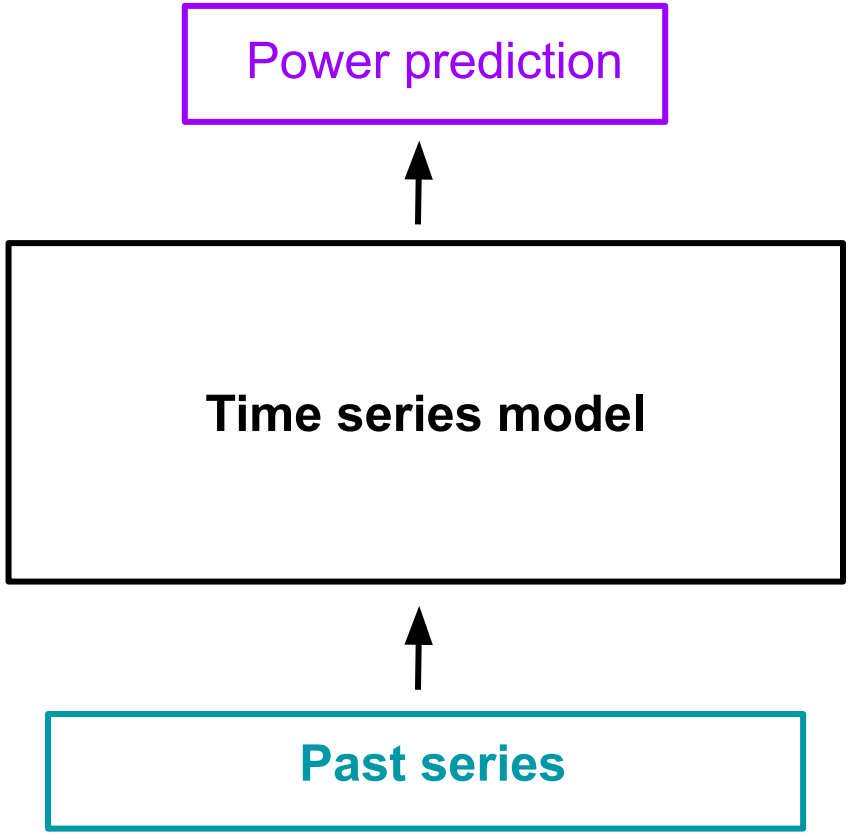
Forecasting

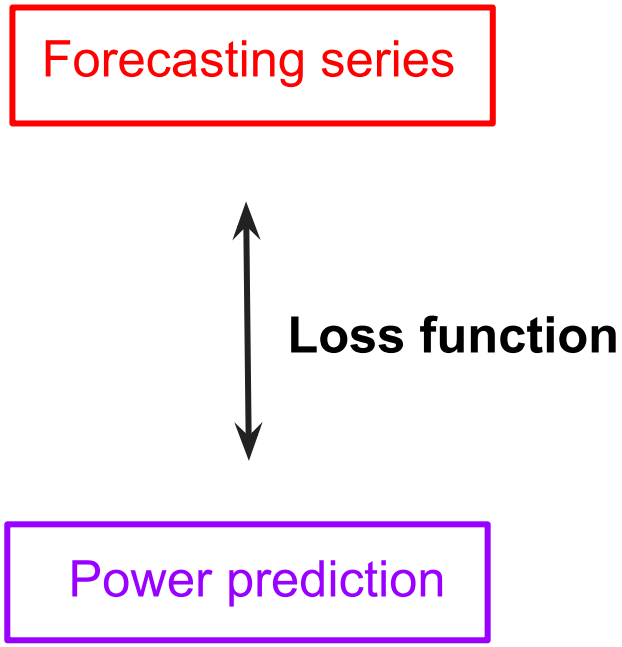
Sliding Window Technique



Forecasting




Power prediction
Time series model
Past series

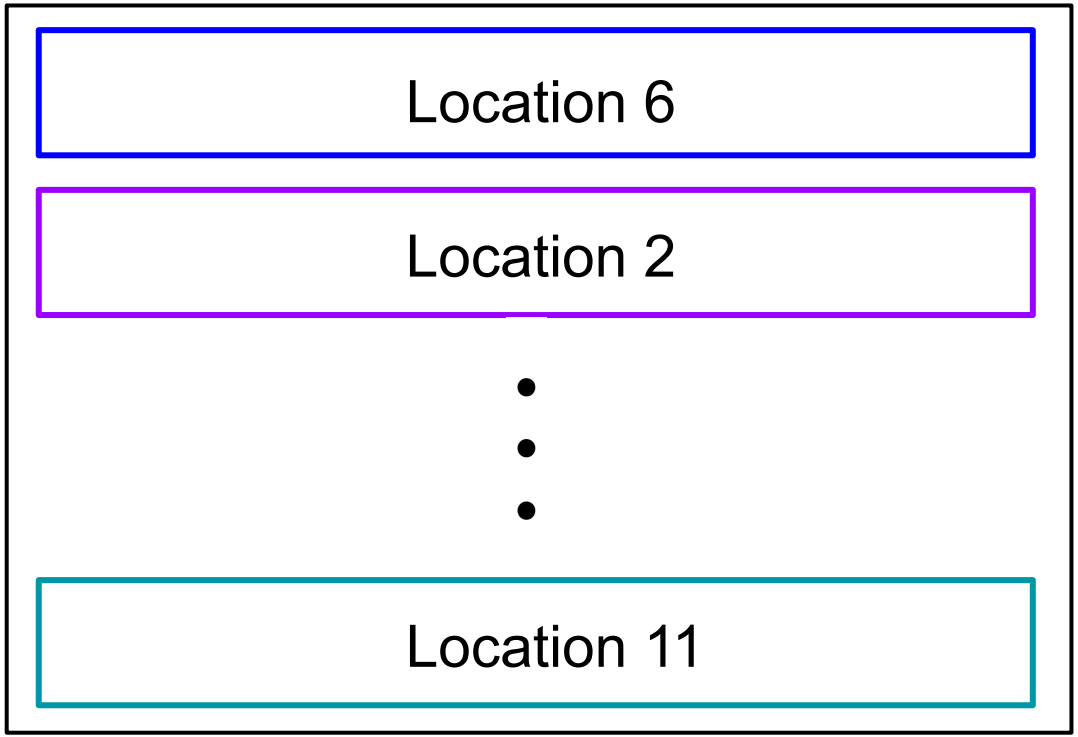

Forecasting series
Loss function
Power prediction

Generate Mini-Batches

- **For all locations:**
  - 80% windows for training
  - 20% windows for validation

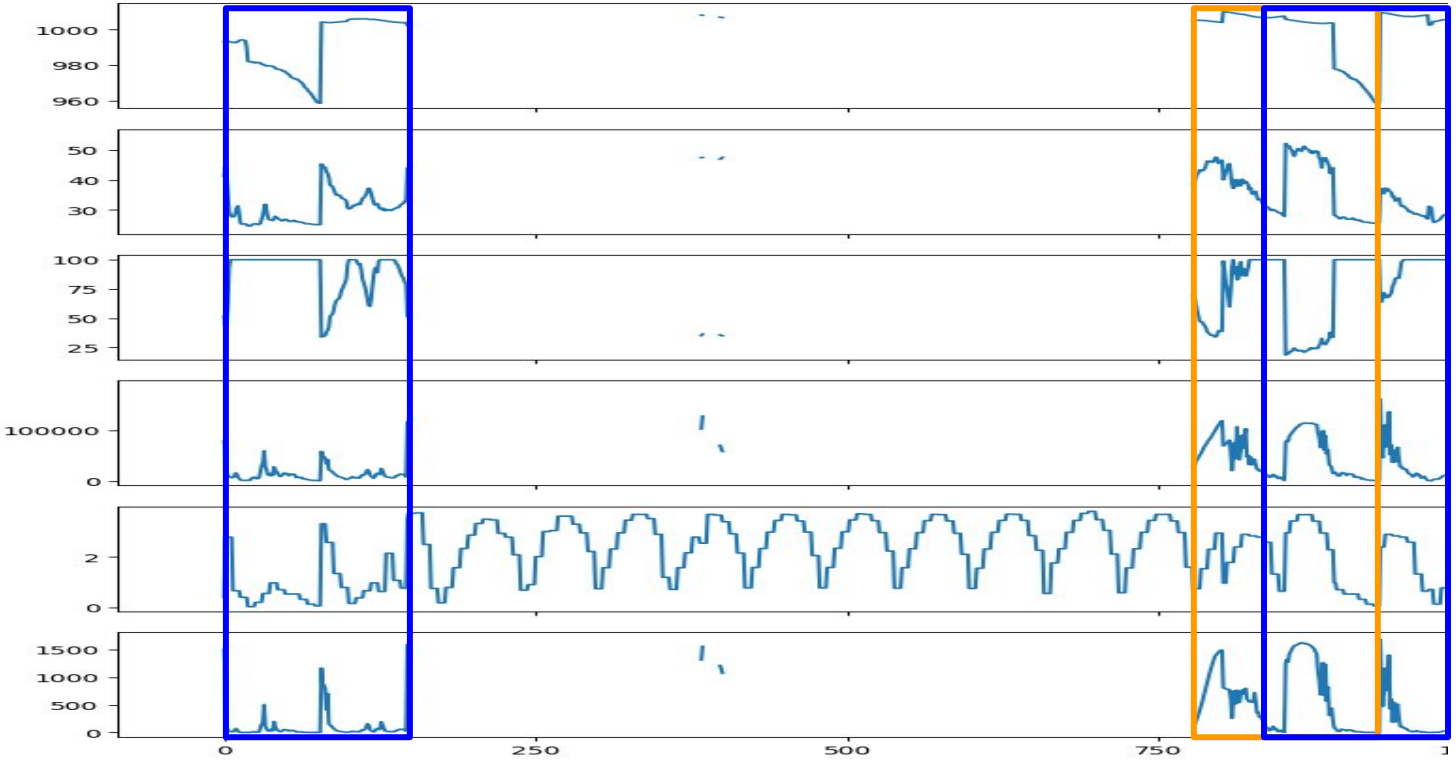
- **Merge all training / validation windows into one**
  - Reason: all locations use same measurement devices
  - Convenience: no need to train the model for each location
  - More data helps improving accuracy and generalizaiton

Generate Mini-Batches

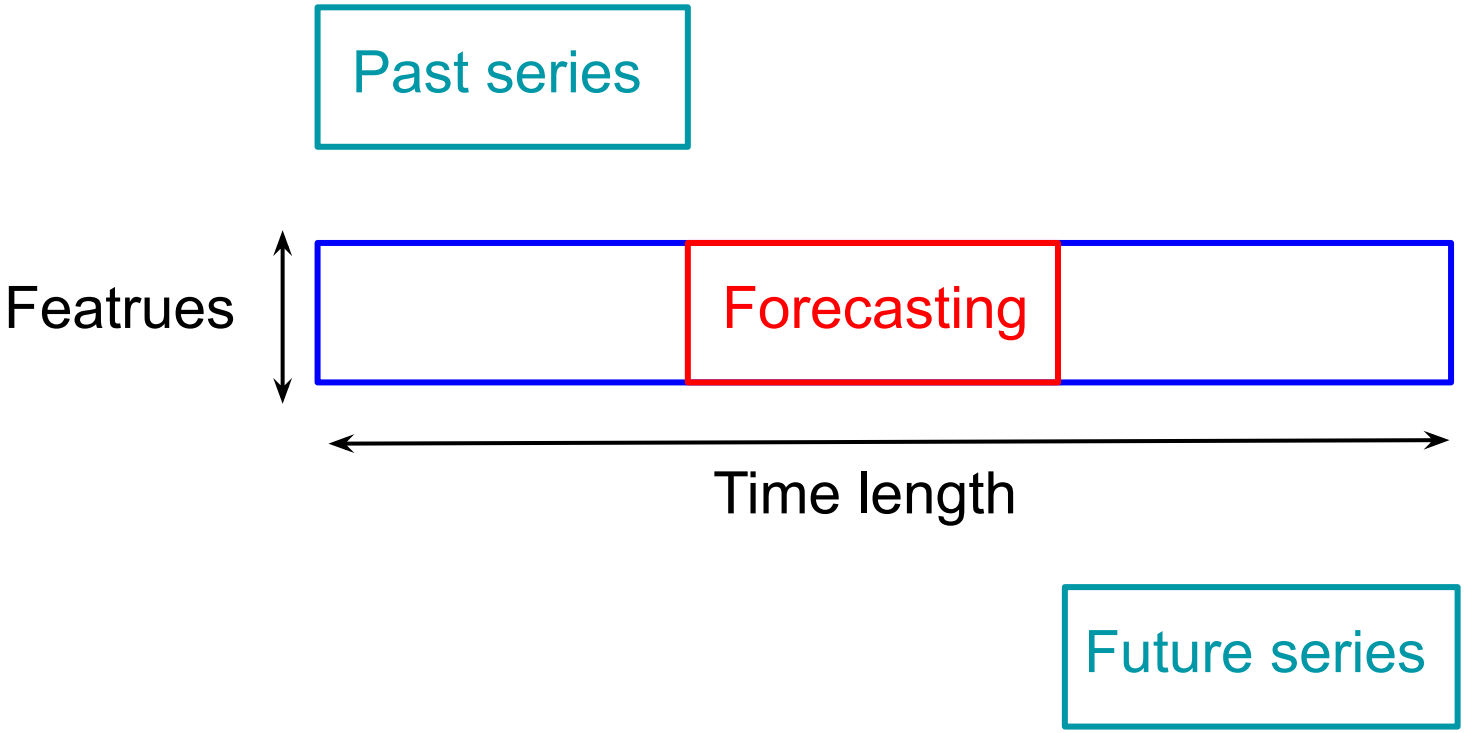


Imputation

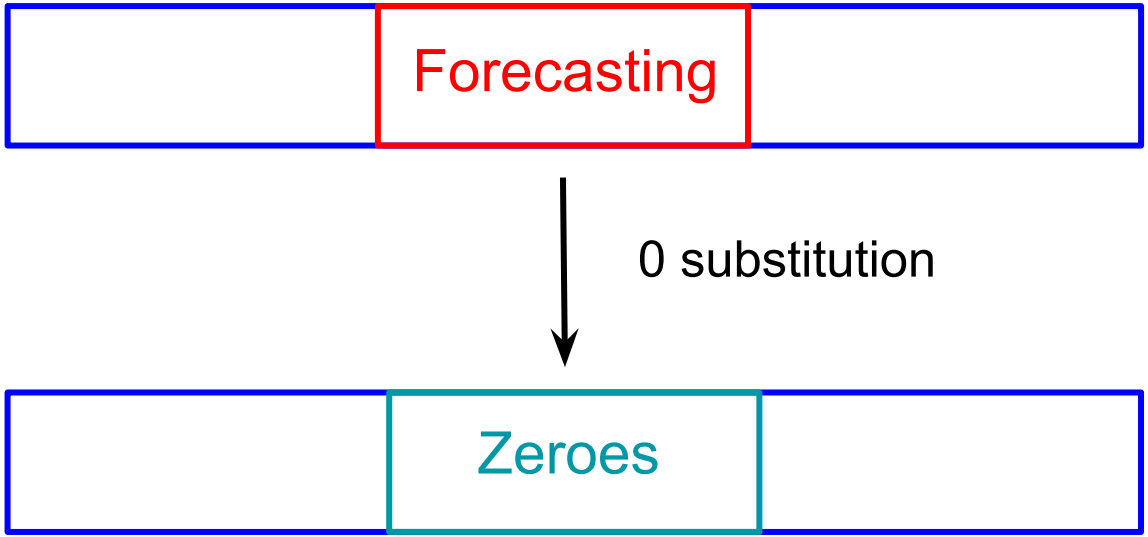
Sliding Window Technique

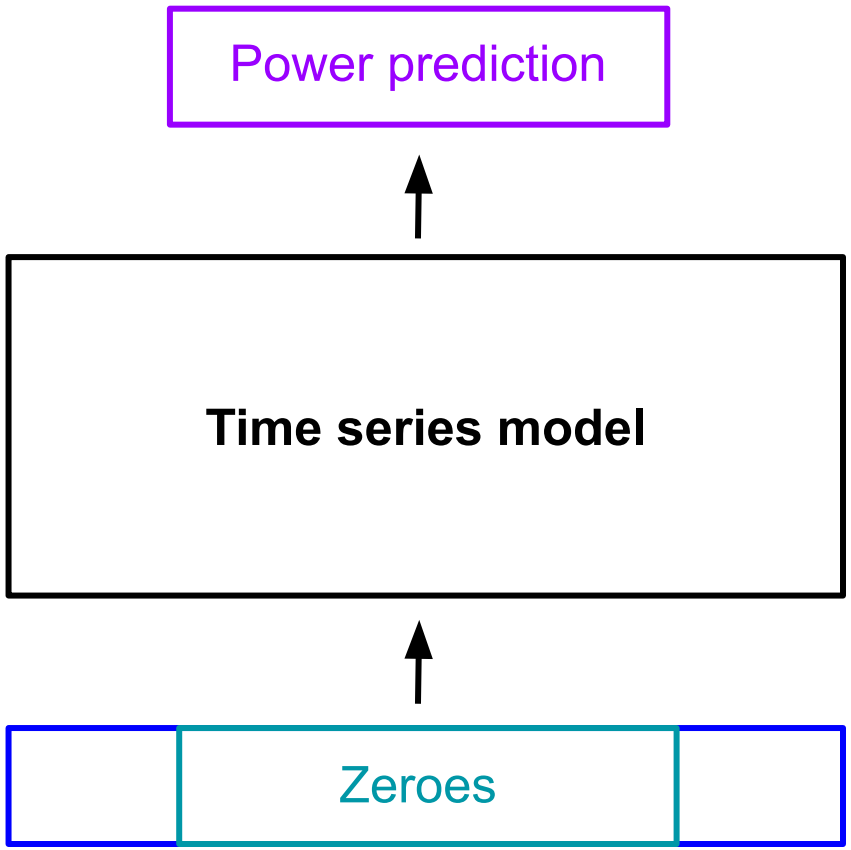


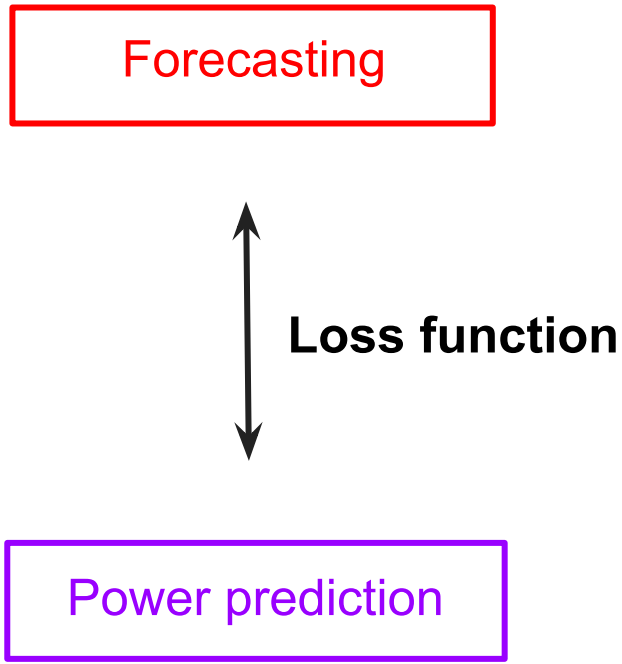
Imputation


Past series
Featrues
Forecasting
Time length
Future series

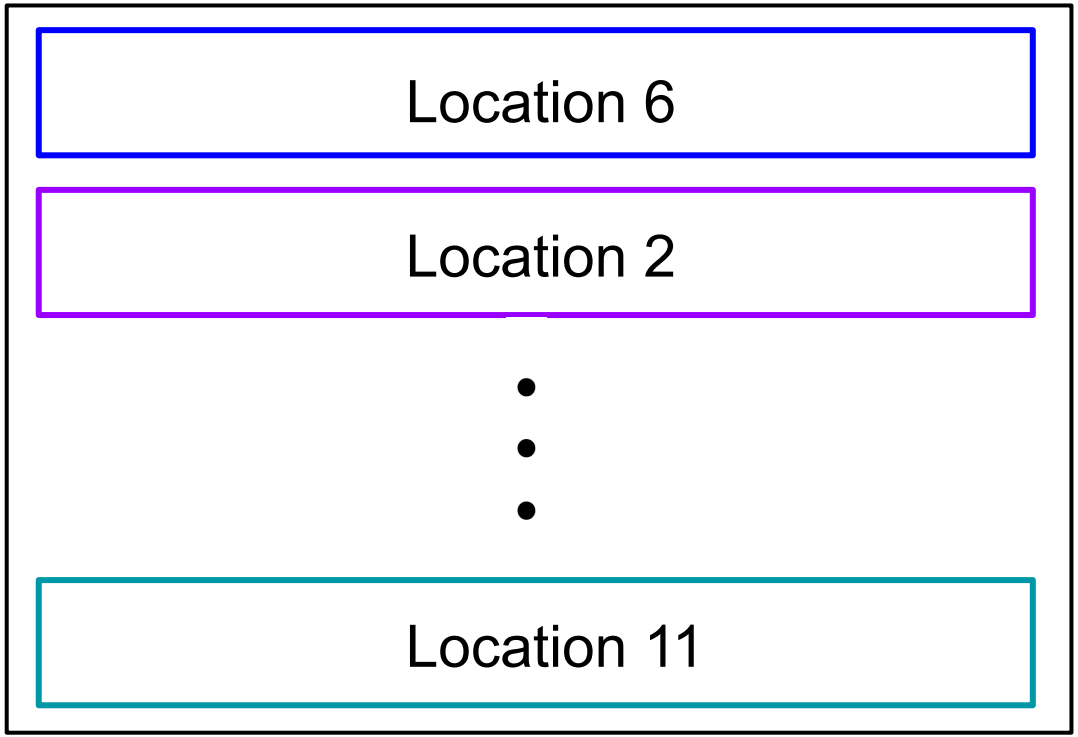
Imputation

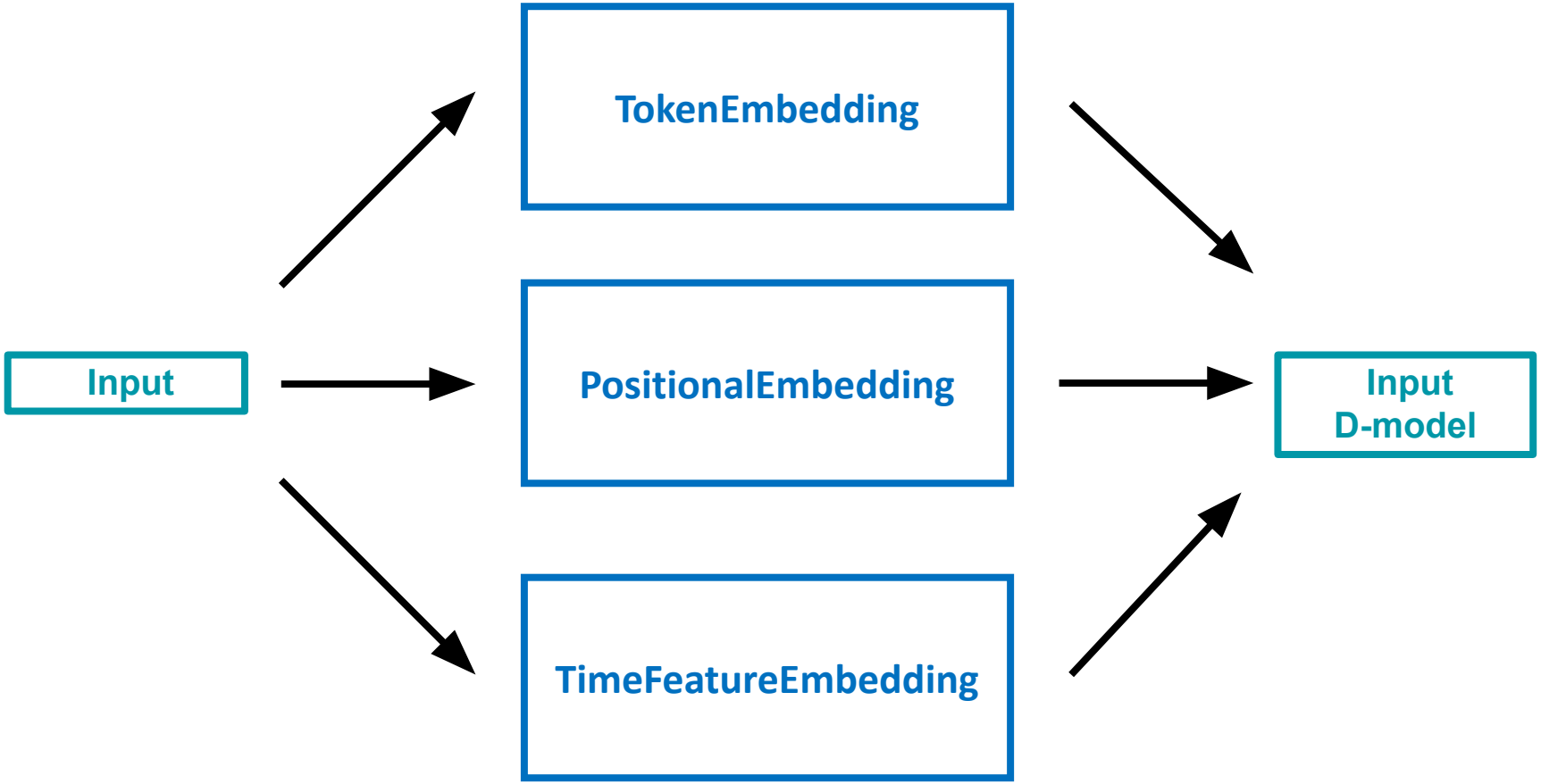


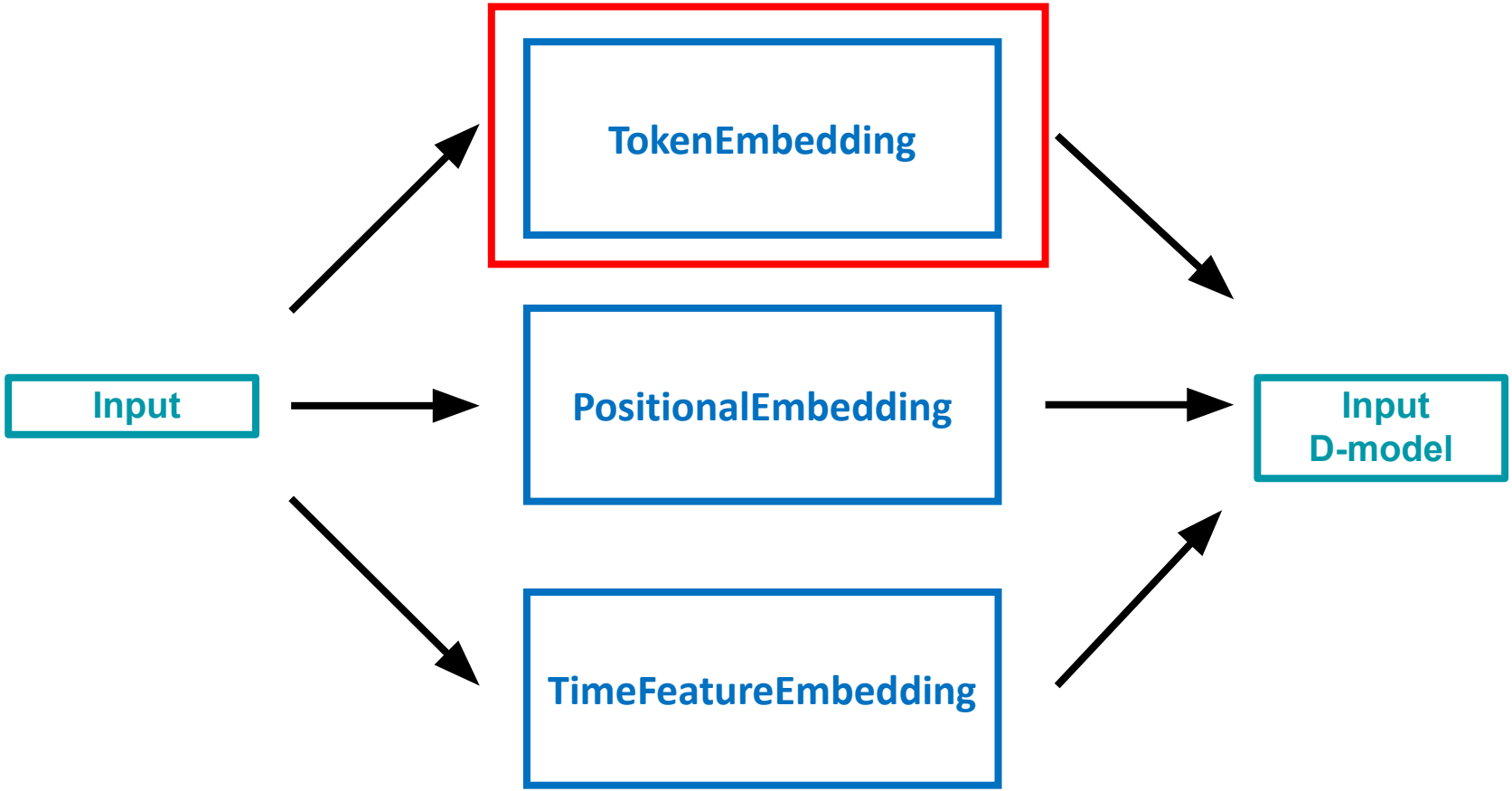

Power prediction
Time series model
Zeroes


Forecasting
Loss function
Power prediction

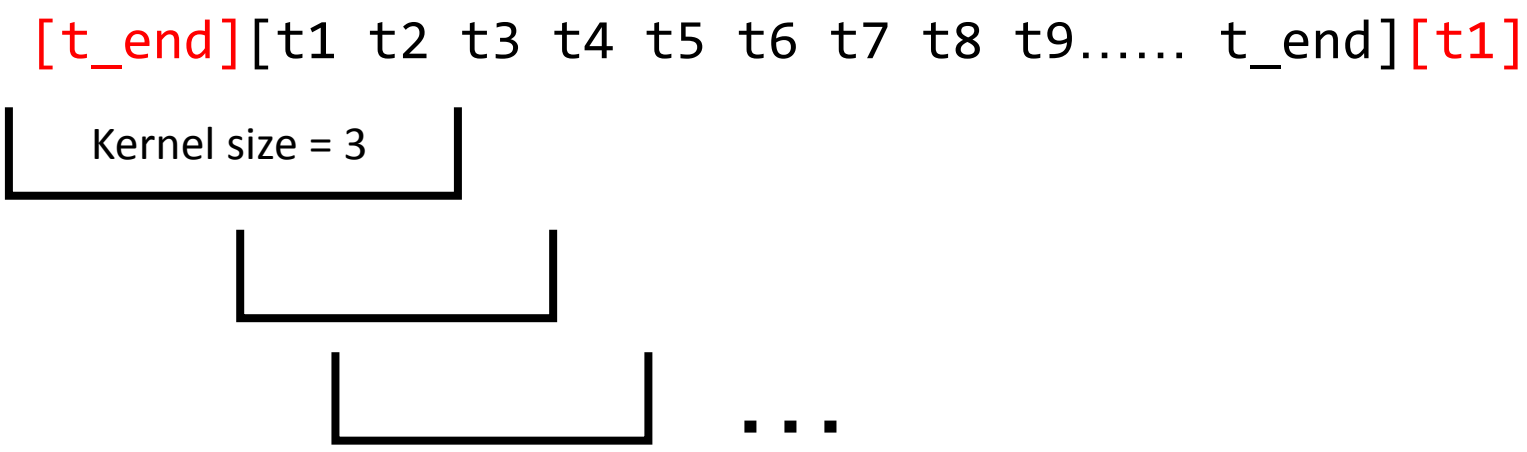
Generate Mini-Batches



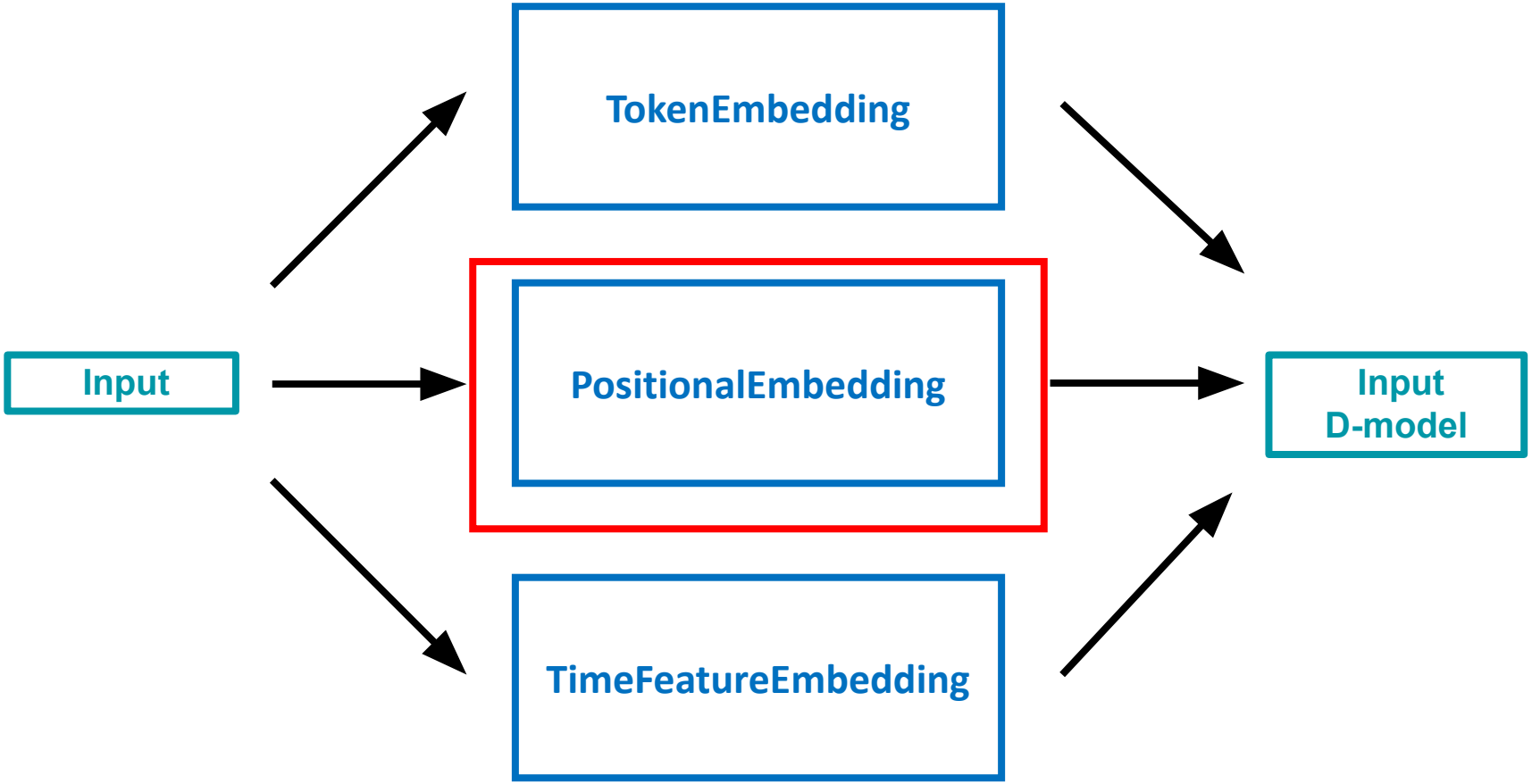

Input
TokenEmbedding
PositionalEmbedding
TimeFeatureEmbedding
Input
D-model


TokenEmbedding
Input
PositionalEmbedding
Input
D-model
TimeFeatureEmbedding

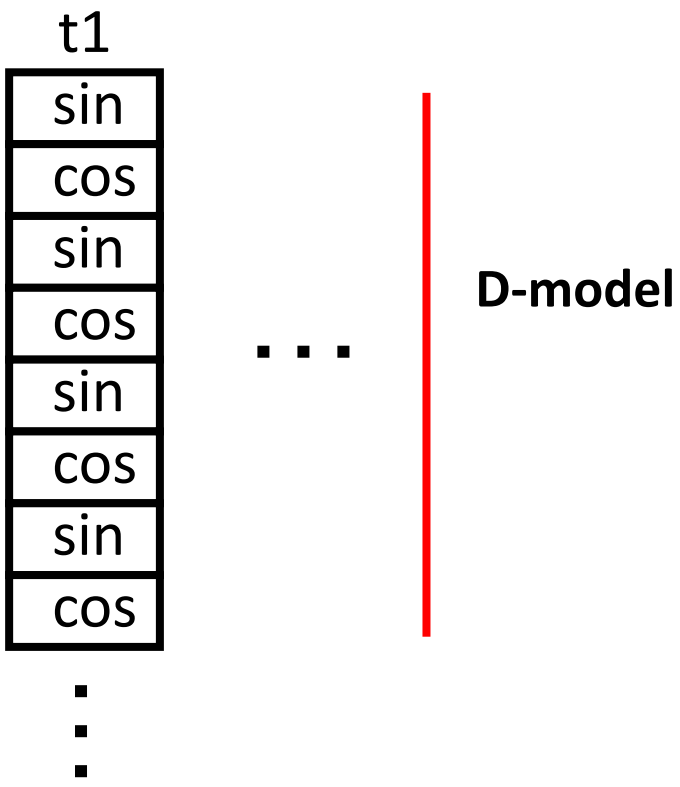
TokenEmbedding

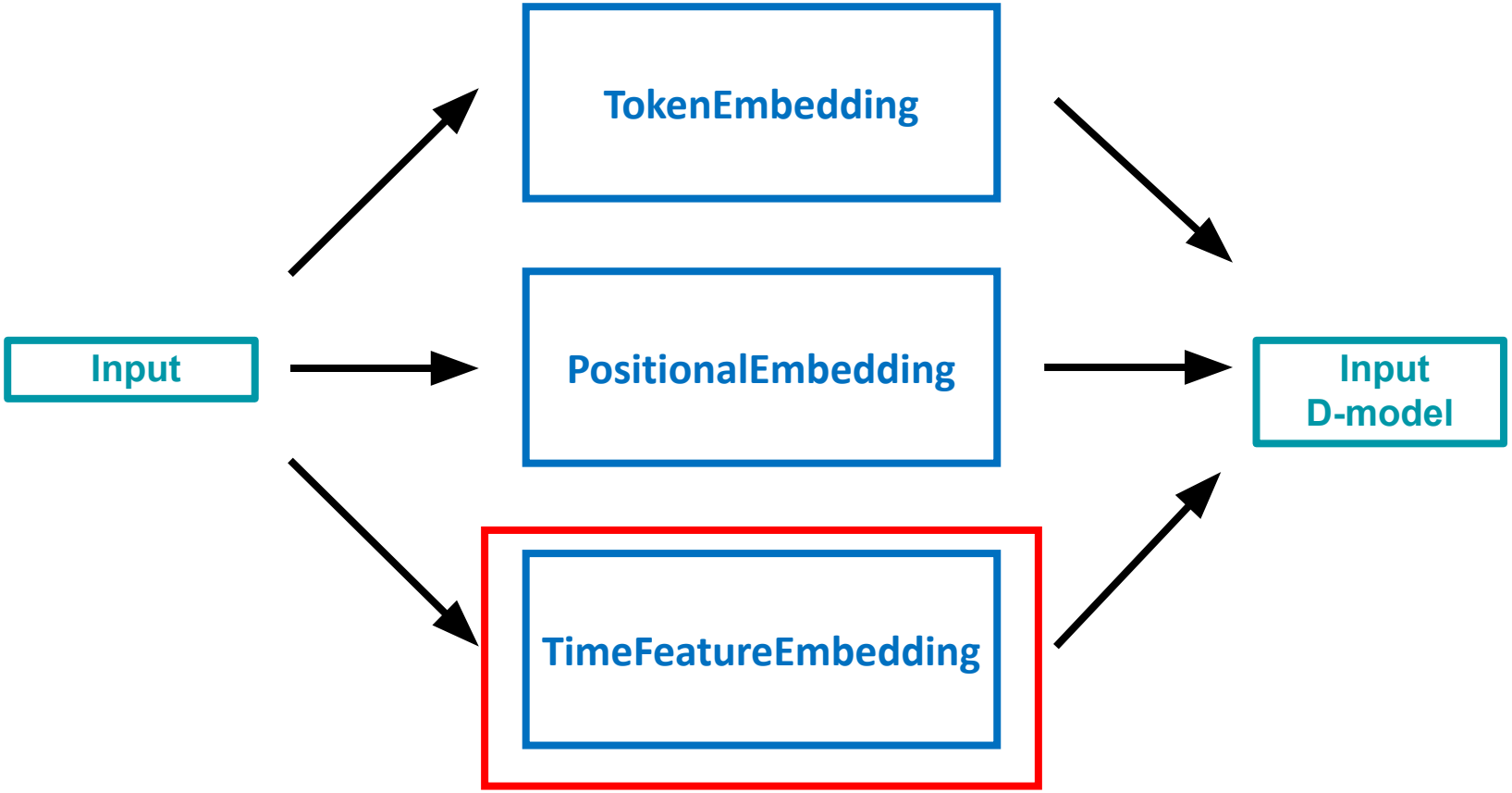


Circular Padding 1D Convolution

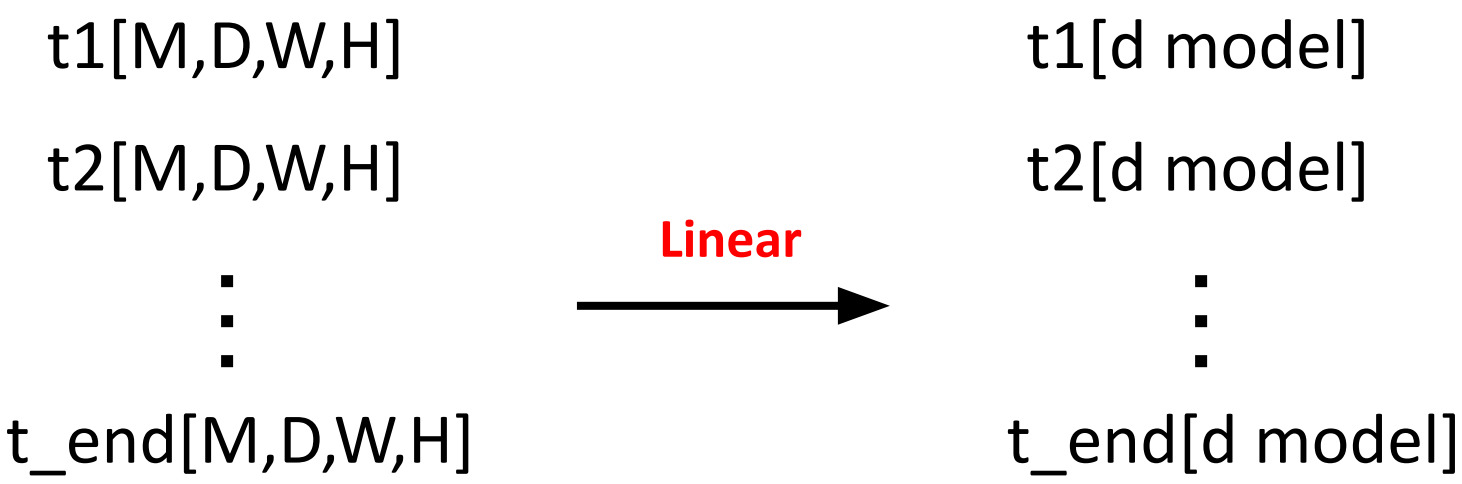

TokenEmbedding
Input
PositionalEmbedding
Input
D-model
TimeFeatureEmbedding

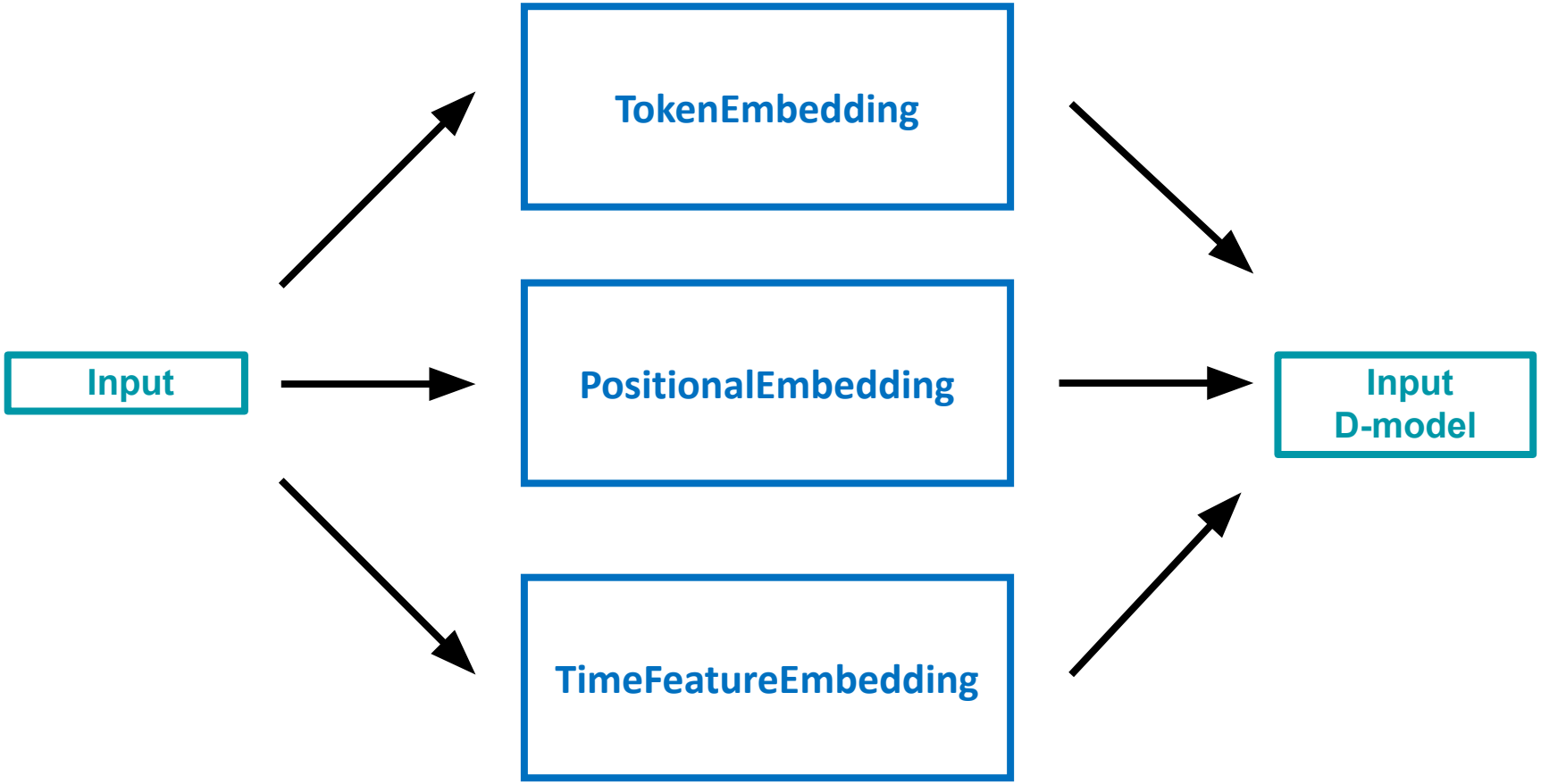
PositionalEmbedding




Input
TokenEmbedding
PositionalEmbedding
TimeFeatureEmbedding
Input
D-model

TimeFeatureEmbedding




Input
TokenEmbedding
PositionalEmbedding
TimeFeatureEmbedding
Input
D-model

Results and Analysis

Hyperparameter Search

Using Optuna ------ based on Bayesian optimization and Tree-Structured Parzen Estimators (TPE).

Trial:50

Forecasting

Parameter	Range
seq_len	2 to 48 (step=2)
d_model	16 to 64 (step=2)
batch_size	[8, 16, 32, 64]
learning_rate	1e-4 to 1e-3 (log scale)
dropout	0.1 to 0.3 (step=0.05)
top_k	[3, 5, 7]

Imputation

Parameter	Range
seq_len	66 to 180 (step=2)
d_model	16 to 64 (step=2)
batch_size	[8, 16, 32, 64]
learning_rate	1e-4 to 1e-3 (log scale)
dropout	0.1 to 0.3 (step=0.05)
top_k	[3, 5, 7]

Results and Analysis

Forecasting

Model	Val loss	Private score
LSTM(Baseline)	0.5172017	*
Timesnet	0.505542	1140101.16
Crossformer	0.5083890	1187184.86
Non stationary Transformer	1.0721987	1742295.59

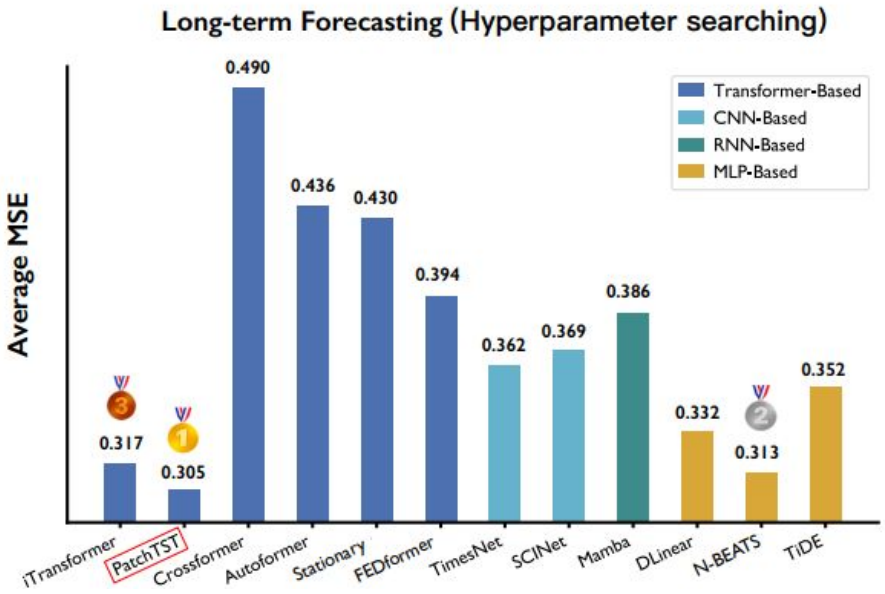
Imputation

Model	Val loss	Private score
LSTM(Baseline)	0.5172017	*
Timesnet	0.22368	849112.73
Crossformer	0.267	1051224.53
PatchTST	0.290510	1163862.5

44	TEAM_6310	3	37	849112.7

44/500 ≈ 9%

Comparison with Literature Results



Forecasting

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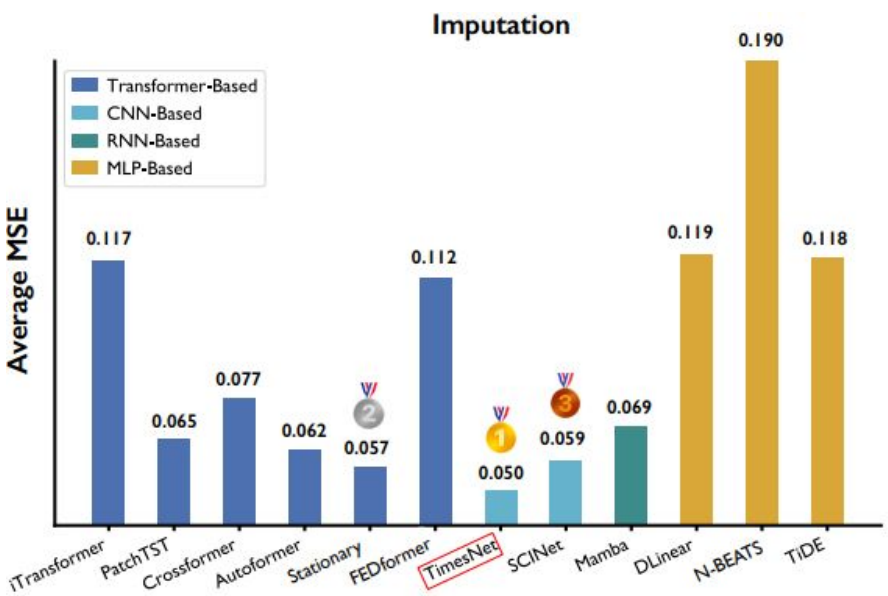

Imputation

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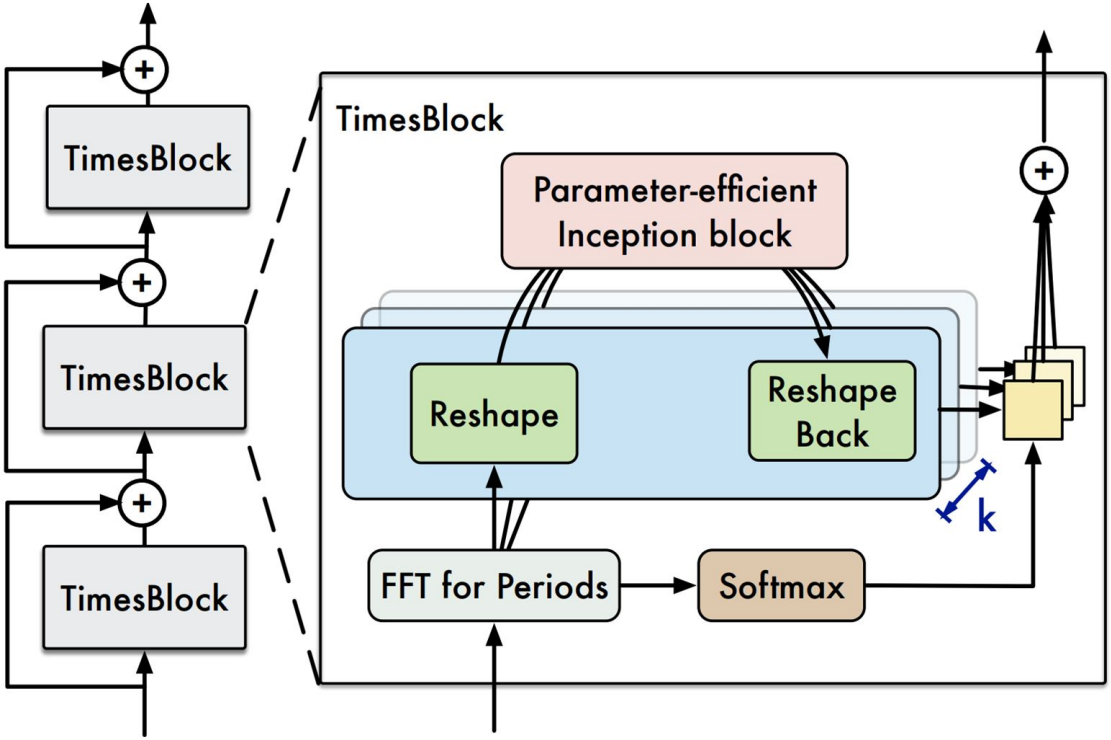
Revisiting TimesNet



Why Does TimesNet Perform the Best?

TimesNet converts time series into a 2D perspective and combines FFT periodic analysis, enabling it to **simultaneously capture both local continuity and periodic patterns** in data, this multi-perspective analysis capability significantly improves the accuracy of imputation tasks.

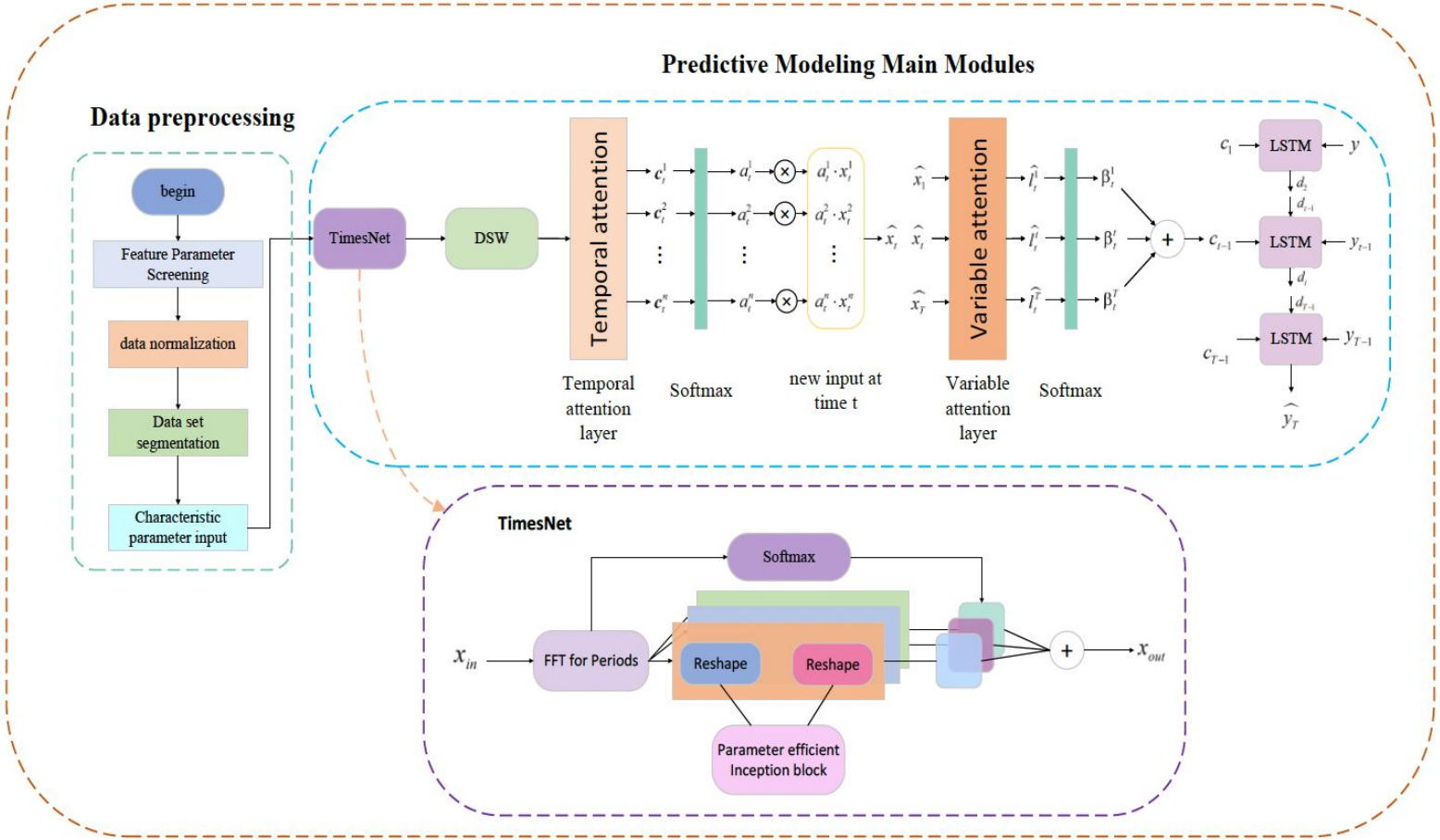
Conclusions

Conclusions

- **Data analysis and feature engineering**
- **Try forecasting and imputation method**
- **Literature review and results comparison**

Conclusions

- **Future work**
  - Higher training epochs
  - Timesnet
    - Change Inception to Resnext
    - Other advanced model based on Timesnet


Predictive Modeling Main Modules
Data preprocessing
begin
Feature Parameter Screening
data normalization
Data set segmentation
Characteristic parameter input
TimesNet
DSW
Temporal attention
Temporal attention layer
Softmax
new input at time t
Variable attention
Variable attention layer
Softmax
LSTM
LSTM
LSTM
TimesNet
Softmax
FFT for Periods
Reshape
Reshape
Parameter efficient Inception block

THANK YOU